

Course Code: CAI2512

Course Name: Neural Network and Fuzzy Logic

ASSIGNMENT-01

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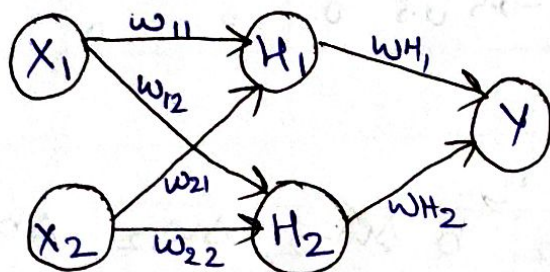
Q. Construct Multilayer Perceptron for XNOR Gate.

XNOR Gate Truth Table :-

A	B	A XNOR B
0	0	1
0	1	0
1	0	0
1	1	1

Input layer: X_1, X_2
Hidden layer: H_1, H_2
Output: Y

Network structure:



Learning rate $\eta = 0.5$

$$H = X_1 w_{11} + X_2 w_{21} + b$$

$$\Delta b = \eta (y - y')$$

$$\Delta w = \eta (y - y') x$$

$$w = w + \Delta w$$

$$b = b + \Delta b$$

Hidden layer (H_1):

Iteration I:

X_1	X_2	y_1	H_1	w_{11}	w_{21}	b	Δw_{11}	Δw_{21}	Δb
0	0	1	0	0	0	0	0	0	0
0	1	0	1	0	0	0.5	0	0.5	-0.5
1	0	0	0	0	0.5	0	0	0	0
1	1	0	1	0	0.5	0	-0.5	-0.5	-0.5

Iteration II:

X_1	X_2	y_1	H_1	w_{11}	w_{21}	b	Δw_{11}	Δw_{21}	Δb
0	0	1	0	-0.5	0	-0.5	0	0	0.5
0	1	0	0	-0.5	0	0	0	0	0
1	0	0	0	-0.5	0	0	0	0	0
1	1	0	0	-0.5	0	0	0	0	0

Iteration III:

x_1	x_2	y_1	H_1	w_{11}	w_{21}	b	Δw_{11}	Δw_{21}	Δb
0	0	1	0	-0.5	0	0	0	0	0.5
0	1	0	1	-0.5	0	0.5	0	-0.5	-0.5
1	0	0	0	-0.5	-0.5	0	0	0	0
1	1	0	0	-0.5	-0.5	0	0	0	0

Iteration IV:

x_1	x_2	y_1	H_1	w_{11}	w_{21}	b	Δw_{11}	Δw_{21}	Δb
0	0	1	0	-0.5	-0.5	0	0	0	0.5
0	1	0	0	-0.5	-0.5	0.5	0	0	0
1	0	0	0	-0.5	-0.5	0.5	0	0	0
1	1	0	0	-0.5	-0.5	0.5	0	0	0

Iteration V:

x_1	x_2	y_1	H_1	w_{11}	w_{21}	b	Δw_{11}	Δw_{21}	Δb
0	0	1	1	-0.5	-0.5	0.5	0	0	0
0	1	0	0	-0.5	-0.5	0.5	0	0	0
1	0	0	0	-0.5	-0.5	0.5	0	0	0
1	1	0	0	-0.5	-0.5	0.5	0	0	0

Hidden Layer (H_2):

Iteration I:

x_1	x_2	y_1	H_2	w_{22}	w_{12}	b	Δw_{22}	Δw_{12}	Δb
0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0.5	0.5	0.5

Iteration II:

x_1	x_2	y_1	H_2	w_{22}	w_{12}	b	Δw_{22}	Δw_{12}	Δb
0	0	0	1	0.5	0.5	0.5	0	0	-0.5
0	1	0	1	0.5	0.5	0	0	-0.5	-0.5
1	0	0	0	0.5	0	-0.5	0	0	0
1	1	1	0	0.5	0	-0.5	0.5	0.5	0.5

Iteration III:

x_1	x_2	y_1	H_2	w_{22}	w_{12}	b	Δw_{22}	Δw_{12}	Δb
0	0	0	0	1	0.5	0	0	0	0
0	1	0	1	1	0.5	0	0	0	0
1	0	0	1	1	0.5	0	0	0	0
1	1	1	1	1	0.5	0	0	0	0

Iteration IV:

x_1	x_2	y_1	H_2	w_{22}	w_{12}	b	Δw_{22}	Δw_{12}	Δb
0	0	0	0	1	0.5	0	0	0	0
0	1	0	1	1	0.5	0	0	-0.5	-0.5
1	0	0	1	1	0	-0.5	-0.5	0	-0.5
1	1	1	0	0.5	0	-1	0.5	0.5	0.5

Iteration V:

x_1	x_2	y_1	H_2	w_{22}	w_{12}	b	Δw_{22}	Δw_{12}	Δb
0	0	0	0	1	0.5	-0.5	0	0	0
0	1	0	0	1	0.5	-0.5	0	-0.5	-0.5
0	0	0	0	1	0	-1	0	0	0
1	1	1	0	1	0	-1	0.5	0.5	0.5

Iteration VI:

x_1	x_2	y_1	H_2	w_{22}	w_{12}	b	Δw_{22}	Δw_{12}	Δb
0	0	0	0	1.5	0.5	-0.5	0	0	0
0	1	0	0	1.5	0.5	-0.5	0	0	0
1	0	0	1	1.5	0.5	-0.5	-0.5	0	-0.5
1	1	1	1	1	0.5	-1	0	0	0

Iteration VII:

x_1	x_2	y_1	H_2	w_{22}	w_{12}	b	Δw_{22}	Δw_{12}	Δb
0	0	0	0	1	0.5	-1	0	0	0
0	1	0	0	1	0.5	-1	0	0	0
1	0	0	0	1	0.5	-1	0	0	0
1	1	1	1	1	0.5	-1	0	0	0

Compute both Hidden layer (H_1 and H_2):

Iteration I:

H_1	H_2	y	y'	w_{H1}	w_{H2}	b	Δw_{H1}	Δw_{H2}	Δb
1	0	1	0	0	0	0	0.5	0	0.5
0	0	0	1	0.5	0	0.5	0	0	-0.5
0	0	0	0	0.5	0	0	0	0	0
0	1	1	0	0.5	0	0	0	0.5	0.5

Iteration II:

H_1	H_2	y	y'	w_{H1}	w_{H2}	b	Δw_{H1}	Δw_{H2}	Δb
1	0	1	1	0.5	0.5	0.5	0	0	0
0	0	0	1	0.5	0.5	0.5	0	0	-0.5
0	0	0	0	0.5	0.5	0	0	0	0
0	1	1	1	0.5	0.5	0	0	0	0

Iteration III:

H_1	H_2	y	y'	w_{H1}	w_{H2}	b	Δw_{H1}	Δw_{H2}	Δb
1	0	1	1	0.5	0.5	0	0	0	0
0	0	0	0	0.5	0.5	0	0	0	0
0	0	0	0	0.5	0.5	0	0	0	0
0	1	1	1	0.5	0.5	0	0	0	0